Omoze Idehenre
4/11/2026
PKI Week 6
Assets

Lab 1

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Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in expired_cert.pem -noout -dates
notBefore=Apr  9 00:00:00 2015 GMT
notAfter=Apr 12 23:59:59 2015 GMT
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in expired_cert.pem -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            4a:e7:95:49:fa:9a:be:3f:10:0f:17:a4:78:e1:69:09
    Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=GB, ST=Greater Manchester, L=Salford, O=COMODO CA Limited, CN=COMODO RSA Domain Validation Secure Server CA
        Validity
            Not Before: Apr  9 00:00:00 2015 GMT
            Not After : Apr 12 23:59:59 2015 GMT
        Subject: OU=Domain Control Validated, OU=PositiveSSL Wildcard, CN=*.badssl.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                RSA Public-Key: (2048 bit)
                Modulus:
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                CA:FALSE
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
            X509v3 Certificate Policies:
                Policy: 1.3.6.1.4.1.6449.1.2.2.7
                  CPS: https://secure.comodo.com/CPS
                Policy: 2.23.140.1.2.1

            X509v3 CRL Distribution Points:

                Full Name:
                  URI:http://crl.comodoca.com/COMODORSADomainValidationSecureServerCA.crl

            Authority Information Access:
                CA Issuers - URI:http://crt.comodoca.com/COMODORSADomainValidationSecureServerCA.crt
                OCSP - URI:http://ocsp.comodoca.com

            X509v3 Subject Alternative Name:
                DNS:*.badssl.com, DNS:badssl.com
    Signature Algorithm: sha256WithRSAEncryption
```

Lab 2

```
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ awk 'BEGIN {c=0} /BEGIN CERTIFIC
ATE/ {c++} c==1 {print}' fullchain.txt > leaf_cert.pem
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in leaf_cert.pem -
noout -subject -issuer -dates
subject= /CN=*.badssl.com
issuer= /C=US/O=Let's Encrypt/CN=R13
notBefore=Mar 24 20:02:52 2026 GMT
notAfter=Jun 22 20:02:51 2026 GMT
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

```
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in leaf_cert.pem -
noout -subject -issuer -dates
subject= /CN=*.badssl.com
issuer= /C=US/O=Let's Encrypt/CN=R13
notBefore=Mar 24 20:02:52 2026 GMT
notAfter=Jun 22 20:02:51 2026 GMT
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify leaf_cert.pem
CN = *.badssl.com
error 20 at 0 depth lookup:unable to get local issuer certificate
leaf_cert.pem: verification failed: 20 (unable to get local issuer certificate)
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in leaf_cert.pem -
noout -text | grep -A3 "Authority Information Access"
            Authority Information Access:
                CA Issuers - URI:http://r13.i.lencr.org/

            X509v3 Subject Alternative Name:
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ curl -s http://r13.i.lencr.org/
-o intermediate.der
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -inform DER -in int
ermediate.der -out issuer_cert.pem
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl verify -untrusted issuer
_cert.pem leaf_cert.pem
leaf_cert.pem: OK
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

Lab 3

```
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl s_client -connect wrong.
host.badssl.com:443 -servername wrong.host.badssl.com -showcerts </dev/null 2>/d
ev/null > san_full.txt
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ awk 'BEGIN {c=0} /BEGIN CERTIFIC
ATE/ {c++} c==1 {print}' san_full.txt > san_cert.pem
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in san_cert.pem -n
oout -subject
subject= /CN=*.badssl.com
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in san_cert.pem -n
oout -text | grep -A5 "Subject Alternative Name"
            X509v3 Subject Alternative Name:
                DNS:*.badssl.com, DNS:badssl.com
```

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    Timeout   : 7200 (sec)
    Verify return code: 0 (ok)
---
closed
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl s_client -connect wrong.
host.badssl.com:443 -servername wrong.host.badssl.com </dev/null 2>&1 | grep "Ve
rify return code"
    Verify return code: 0 (ok)
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in san_cert.pem -n
oout -text | grep -A5 "Subject Alternative Name"
            X509v3 Subject Alternative Name:
                DNS:*.badssl.com, DNS:badssl.com
            X509v3 Certificate Policies:
                Policy: 2.23.140.1.2.1

            X509v3 CRL Distribution Points:
Mac:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```